
BAR: Budgeted Actor Refinement Decouples Bootstrap Exposure from Policy Reach

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Abstract

TD3+BC uses one actor branch both to construct bootstrapped critic targets and to produce the deployed policy, forcing bootstrap exposure and policy reach to move together. We introduce budgeted actor refinement (BAR), a persistent K -actor chain that separates these roles: a Polyak copy of the first actor supplies TD-target actions, while the final actor is deployed. A finite-step Lipschitz bound shows that one-step target perturbation depends on first-hop displacement independently of later refinement reach; an ideal Wasserstein-2 proximal analysis characterizes the complementary local geometry. Across two disjoint, host-separated seed pairs on nine D4RL locomotion tasks and fourteen coefficient budgets, the four-seed, equally task-and-budget-weighted proximal aggregate over $T \geq 4$ is strictly ordered by depth: the mean rises from 25.38 at $K = 1$ to 63.97 at $K = 4$, while score- < 20 seed-mean cells fall from 34/63 to 8/63. Each seed pair separately reproduces the full $K = 1 < 2 < 3 < 4$ aggregate ordering. The $K = 4$ minus $K = 3$ gain is 12.06 with a task-resampling interval of [3.84, 21.11]; its .34 cellwise median and eight rescued cells show that low-score rescue drives most of the added-depth aggregate gain. In a 270-method-checkpoint TD3/P2/P3 audit, three-hop refinement lowers sample-anchored next-state target-action displacement in 88/90 matched comparisons, while the typical final-policy proxy does not contract. These results establish refinement depth as a consequential stability control and identify finite-step bootstrap-exposure

decoupling as BAR’s central empirical mechanism signature.

1 INTRODUCTION

Offline reinforcement learning (RL) seeks a policy from a fixed dataset without additional environment interaction. Its central difficulty is not merely to select a high-value action, but to determine how far policy improvement can proceed before the learned critic becomes unreliable. TD3+BC regularizes a deterministic actor toward dataset actions while using a Polyak copy of that actor in bootstrapped critic targets (Fujimoto and Gu, 2021). The same policy branch therefore determines both the action shift entering critic training and the reach of the evaluated policy. Aggressive improvement can move target actions into regions where critic error is amplified, whereas conservative improvement can leave useful policy improvement unrealized.

We study this coupling through *budgeted actor refinement* (BAR), a re-centered actor architecture for TD3+BC. BAR represents a nominal improvement budget with K sequential actor updates. The first actor is anchored to dataset actions, and each later actor is anchored to the stop-gradient output of its predecessor. A target copy of the first refined actor supplies the next actions in TD targets, while the final actor is used for evaluation. The architecture thus assigns critic-target construction and deployment to distinct policy branches without changing the critic update rule.

Let D_{critic} denote the squared displacement of the critic-coupled first-hop branch and let D_{final} denote that of the evaluated final policy. A one-hop actor identifies these quantities; BAR assigns them to μ_1 and μ_K , respectively. This separation can increase final-policy reach at comparable first-hop movement or contract both movements near an unstable training outcome. It also introduces a countervailing mismatch because the critic target follows the μ_1 branch while evaluation follows μ_K . The method therefore exposes

a design axis between target-policy displacement and deployment reach rather than guaranteeing that either quantity is independently optimal.

For deterministic actors, the squared action anchor in TD3+BC is a Wasserstein-2 movement cost in the Dirac-policy limit. Under a fixed critic, inactive action projection, accurate subproblem solutions, and small *realized* steps, explicit and implicit updates are first-order discretizations of the same Wasserstein gradient flow. At finite steps, a proximal comparison yields an improvement margin only relative to the learned critic. At large nominal budgets, projection, neural actor fitting, critic drift, saturation, and the seeded optimization trajectory determine the observed end-to-end outcome. We keep these local, finite-step, and coupled regimes distinct throughout the analysis.

We evaluate this formulation with a complete fixed-budget phase diagram over nine D4RL locomotion tasks, fourteen budgets, and four proximal seeds. The equally task-and-budget-weighted high-budget mean is strictly ordered $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$ for $K = 1, 2, 3, 4$, while low-return cells fall $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$. Both disjoint host-separated seed pairs reproduce the full aggregate ordering. The $K = 4$ minus $K = 3$ gain is large but highly nonuniform: eight rescued cells account for 54.7% of the total added-depth gain over the 63 high-budget cells. An action-metric-matched projected-linearized realization improves in the same direction. Direct next-state diagnostics show that three-hop BAR lowers sample-anchored target-action displacement in 88/90 matched comparisons with TD3+BC, while its current-state final-policy displacement proxy is lower in only 36/90. Low-return runs additionally exhibit orders-of-magnitude increases in TD-error tails, larger realized movements, and critic gains that reverse sign under a controlled common critic.

Our contributions are as follows.

- We formulate BAR for TD3+BC as a re-centered actor composition that assigns target-policy displacement and final-policy reach to distinct branches.
- We give a Wasserstein proximal interpretation, derive an inexact finite-step margin with an explicit critic-error term, bound first-hop TD-target perturbation, and separate these statements from the fixed-critic small-step flow limit.
- Through four-seed proximal sweeps up to $K = 4$, a two-seed linearized $K = 3$ sweep, direct target-action measurements, critic-failure diagnostics, and targeted controls, we show that refinement depth broadens the empirical large-budget stabil-

ity envelope primarily by rescuing lower-tail failures.

2 RELATED WORK

Behavior-regularized offline actor-critic learning. Offline actor-critic methods must control value extrapolation when a learned policy selects actions outside the fixed dataset. Prior work studies this failure through constrained policy improvement, conservative value learning, and restrictions on actions used in Bellman backups (Fujimoto et al., 2019; Kumar et al., 2019). TD3+BC provides a particularly minimal control: a deterministic actor maximizes a normalized critic while a squared cloning term anchors its actions to the dataset (Fujimoto and Gu, 2021). We keep this critic update and actor family fixed and study which refined actor branch supplies the TD target. One-step offline RL instead limits repeated policy optimization against an off-policy value estimate (Brandfonbrener et al., 2021). BAR retains iterative actor-critic training but distributes a nominal improvement coefficient across a routed actor chain.

Policy geometry and proximal refinement. Trust-region policy optimization is commonly derived from information geometry (Amari and Nagaoka, 2000; Schulman et al., 2015). Optimal-transport geometry gives a different notion of movement: policy optimization can be expressed as a Wasserstein gradient flow, with related natural-gradient and trust-region constructions for continuous control (Zhang et al., 2018; Moskovitz et al., 2020; Terpin et al., 2022; Song et al., 2023). Wasserstein Policy Optimization projects Wasserstein policy flow into a parametric actor class using value-function action gradients (Pfau et al., 2025). For deterministic policies, the statewise Wasserstein-2 cost reduces to squared action displacement. We use that identity to interpret the TD3+BC anchor as a proximal movement cost, while separating the finite-step variational statement from its small-step flow limit.

Target networks and actor-critic coupling. Bootstrapping error can feed critic extrapolation back into actor updates (Fujimoto et al., 2019; Kumar et al., 2019). Polyak averaging delays this feedback but ordinarily copies the same actor branch that will be deployed. Target-deployment routed refinement makes a narrower architectural change: the target copy follows the first local refinement, whereas evaluation uses the final refinement. Our target-perturbation bound and direct displacement audit therefore concern branch assignment inside TD3+BC rather than a new critic objective or a general claim about offline policy extrac-

tion.

3 PRELIMINARIES

We consider a discounted Markov decision process $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ with continuous actions and a fixed offline dataset $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}_{i=1}^N$. A clipped double critic is trained from targets

$$y_\rho = r + \gamma \min_j Q_j^-(s', \rho(s')), \quad (1)$$

where ρ is a target policy. When $\rho(s')$ leaves dataset support, critic error enters the regression target and can be amplified by later actor updates (Fujimoto et al., 2019; Kumar et al., 2019).

For a deterministic actor μ_θ , normalized TD3+BC minimizes

$$\mathcal{L}_{\text{actor}}(\theta) = -\alpha \mathbb{E}_s \bar{Q}(s, \mu_\theta(s)) + \mathbb{E}_{(s,a) \sim \mathcal{D}} \frac{\|\mu_\theta(s) - a\|_2^2}{d}, \quad (2)$$

where $\bar{Q} = \hat{Q}/C$ and $C > 0$ is a stop-gradient critic scale (Fujimoto and Gu, 2021). Ordinarily, a Polyak copy of this actor enters Eq. (1), while the online actor is deployed. Target-policy movement and deployment reach therefore follow the same branch.

A metric proximal update with step $h > 0$ takes the form

$$\pi^+ \in \arg \min_{\pi} \left\{ \hat{\mathcal{E}}(\pi) + \frac{1}{2h} d_{\mathcal{M}}^2(\pi, \pi^{\text{ref}}) \right\}. \quad (3)$$

This minimizing-movement construction is associated with metric gradient flows (Ambrosio et al., 2005; Otto, 2001). For deterministic state-conditioned policies,

$$W_2^2(\delta_{\mu(s)}, \delta_{\nu(s)}) = \|\mu(s) - \nu(s)\|_2^2. \quad (4)$$

Thus the TD3+BC anchor is a Wasserstein-2 movement cost up to fixed scaling. We distinguish its finite-step variational meaning from the small-step, fixed-critic gradient-flow limit and from large-budget coupled training.

4 BUDGETED ACTOR REFINEMENT

At each delayed actor update, BAR maintains K independently initialized, persistent deterministic actors with separate optimizer states. The first actor takes one TD3+BC update toward the sampled dataset actions; each later actor takes one update toward the stop-gradient output of its predecessor on the same minibatch. The online critic is fixed within this chain; we denote its first Q-function by $\hat{Q}$. A Polyak copy

of the first actor supplies TD-target actions, and the final actor is evaluated. We first derive the proximal operator underlying these updates and then formalize their composition.

4.1 Offline actor updates as proximal improvement

Let $\rho_{\mathcal{D}}$ be the empirical state distribution and consider a deterministic actor $\mu : \mathcal{S} \rightarrow \mathcal{A} \subset \mathbb{R}^d$. At one training iteration, the critic parameters are held fixed while the actor chain is updated. We write the current critic as $\hat{Q}_t$, use $\epsilon = 10^{-6}$, and define the implementation-level proximal scales by

$$\begin{aligned} C_{1,t}^P &= \text{sg} \left(\mathbb{E}_s |\hat{Q}_t(s, \mu_{1,t}^{\text{pre}}(s))| + \epsilon \right), \\ C_{k,t}^P &= \text{sg} \left(\mathbb{E}_s |\hat{Q}_t(s, \text{sg} \mu_{k-1,t}^+(s))| + \epsilon \right), \quad k \geq 2. \end{aligned} \quad (5)$$

The normalized critic is $\bar{Q}_{k,t}(s, a) = \hat{Q}_t(s, a)/C_{k,t}^P$. Here *pre* and the superscript $+$ denote actions before and after the current actor update, and *sg* denotes stop-gradient. We suppress t and the superscript when discussing the ideal operator. A fixed positive C only rescales gradient-flow time; recomputing C_k changes the finite-step schedule. Our finite-step statements condition on the realized scales.

For deterministic policies and a state distribution ρ , define the action-averaged Wasserstein metric

$$\mathbf{d}_{d,\rho}^2(\mu, \nu) := \mathbb{E}_{s \sim \rho} \left[\frac{1}{d} \|\mu(s) - \nu(s)\|_2^2 \right]. \quad (6)$$

It is d^{-1} times the state-averaged squared W_2 distance between state-conditioned Dirac policies. We abbreviate $\mathbf{d}_d = \mathbf{d}_{d,\rho_{\mathcal{D}}}$ on the actor-training state distribution. The population counterpart of squared behavior cloning also admits a deterministic reference. Let P_B be the data-generating behavior distribution, let ρ_B be its state marginal, and define $b(s) = \mathbb{E}_{P_B}[a \mid s]$. Then

$$\mathbb{E}_{(s,a) \sim P_B} \frac{1}{d} \|\mu(s) - a\|_2^2 = \mathbf{d}_{d,\rho_B}^2(\mu, b) + \text{const}, \quad (7)$$

where the constant is the conditional action variance and does not depend on μ . The empirical actor loss is a Monte Carlo estimate of the left-hand side, so samplewise squared anchoring targets the conditional-mean behavior map at the population level.

Given a reference policy ν and step $h > 0$, define the proximal refinement operator by

$$\mathcal{P}_h^{\bar{Q}}(\nu) \in \arg \min_{\mu \in \mathcal{M}} \left\{ -\mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \frac{1}{2h} \mathbf{d}_d^2(\mu, \nu) \right\}. \quad (8)$$

Multiplying by $2h$ gives the practical actor objective

$$\mathcal{L}_{h,k}(\mu \mid \nu) = -2h \mathbb{E}_s \bar{Q}_k(s, \mu(s)) + \mathbf{d}_d^2(\mu, \nu). \quad (9)$$

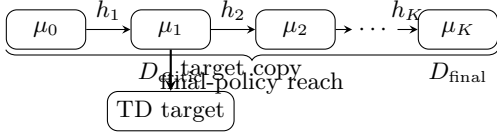


Figure 1: BAR re-centers each actor update. A Polyak copy of μ_1 supplies TD-target actions, while later actors increase or reshape final-policy reach without entering the target policy.

Proposition 1 (Behavior anchoring is Wasserstein proximal improvement). *For deterministic policies, the normalized anchored objective*

$$-\alpha \mathbb{E}_s \bar{Q}(s, \mu(s)) + d_d^2(\mu, \nu)$$

is Eq. (8), up to multiplication by a positive constant, with $h = \alpha/2$. Equivalently, the raw-critic objective $-\lambda \mathbb{E}_s \hat{Q}(s, \mu(s)) + d_d^2(\mu, \nu)$ has $h = \lambda C/2$ when C is fixed. The population counterpart of squared dataset anchoring uses $\nu = b$ by Eq. (7).

The proposition identifies an update operator relative to the *estimated* critic; true-return improvement lies outside the scope of this certificate.

4.2 Re-centered composition and actor-critic dataflow

For population analysis, let $\mu_0 := b$ denote the conditional-mean behavior map and let $h_{1:K}$ be positive step sizes. The idealized BAR chain is

$$\mu_k \in \mathcal{P}_{h_k}^{\hat{Q}}(\text{sg } \mu_{k-1}), \quad k = 1, \dots, K. \quad (10)$$

The implementation approximates each subproblem with one Adam update. Its first actor μ_1 uses sampled dataset actions as its reference; later actors $\mu_2, \dots, \mu_K$ use the preceding online actor under the same current critic. Only the Polyak copy of μ_1 enters the TD target, while μ_K supplies the evaluated action. Because the later persistent actors are independently initialized, their anchoring gradient need not vanish even at $h = 0$ when predecessor and current actor differ. The ideal proximal map therefore motivates, but does not certify, the live one-Adam-step chain.

Our fixed-budget schedule uses the implementation parameter $T = \tau = \alpha/2$, sets $h_k = T/K$, and holds T fixed. Thus $K = 1$ recovers the TD3+BC coefficient $\alpha = 2T$; T is neither a compute budget nor a realized distance constraint. Increasing K jointly changes local coefficient, re-centering frequency, actor count and compute, normalization trajectory, and target/deployment dataflow. Each K is therefore a complete randomized procedure, and its seed average

estimates marginal end-to-end performance. Identical critic initialization across depths is a common-random-number variance-reduction design, not a validity condition.

On the next-state distribution ρ'_D used by the TD target, define the action-averaged population displacements

$$\begin{aligned} D_{\text{critic}} &:= d_{d, \rho'_D}^2(\mu_1, \mu_0), \\ D_{\text{final}} &:= d_{d, \rho'_D}^2(\mu_K, \mu_0). \end{aligned} \quad (11)$$

A one-step actor has $D_{\text{critic}} = D_{\text{final}}$; BAR assigns the two population quantities to distinct policy branches. On fixed validation batches, we measure the sample-anchored online branches and the deterministic next-state Polyak target branch:

$$\begin{aligned} \hat{D}_{\text{critic}} &= \mathbb{E}_i \|\mu_1(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{final}} &= \mathbb{E}_i \|\mu_K(s_i) - a_i\|_2^2/d, \\ \hat{D}_{\text{target}} &= \mathbb{E}_i \|\mu_1^-(s'_i) - a_{i+1}\|_2^2/d. \end{aligned} \quad (12)$$

The direct audit retains only bootstrapped transition pairs; a smoothed version uses common TD3 target-noise draws. Sample anchoring adds a common conditional-action-variance floor in population differences. These are movement measurements, not critic-reliability estimates.

4.3 Finite-step statements

The following statements do not require a small-step Taylor expansion.

Proposition 2 (Inexact estimated-energy margin). *Suppose μ_{k-1} is feasible for the k th actor class and the computed actor satisfies*

$$\mathcal{L}_{h_k, k}(\mu_k \mid \mu_{k-1}) \leq \mathcal{L}_{h_k, k}(\mu_{k-1} \mid \mu_{k-1}) + \varepsilon_k.$$

Then

$$\begin{aligned} m_k^{\hat{Q}} &:= \frac{2h_k}{C_k} \mathbb{E}_s [\hat{Q}(s, \mu_k(s)) - \hat{Q}(s, \mu_{k-1}(s))] \\ &\quad - d_d^2(\mu_k, \mu_{k-1}) \geq -\varepsilon_k. \end{aligned} \quad (13)$$

For the first sample-anchored hop, a statewise dataset action is not generally a feasible deterministic-policy comparator; the population reference in Eq. (7) or a behavior actor is required for a literal certificate. For re-centered hops, the preceding actor is feasible by construction.

The margin is relative to $\hat{Q}$. Let a trusted objective be $Q^\dagger$ and write $\hat{Q} = Q^\dagger + e$. Then

$$\begin{aligned} m_k^{Q^\dagger} &= m_k^{\hat{Q}} - \frac{2h_k}{C_k} \mathbb{E}_s [e(s, \mu_k(s)) \\ &\quad - e(s, \mu_{k-1}(s))]. \end{aligned} \quad (14)$$

Thus estimated-energy descent transfers to $Q^\dagger$ only when critic error is controlled along the realized path. Equation (13) certifies the learned objective rather than monotone true return.

Proposition 3 (First-hop TD-target perturbation bound). *For clarity, consider a noiseless TD target using μ_1 , $y_{\mu_1} = r + \gamma \min_j Q_j^-(s', \mu_1(s'))$, and suppose each target critic $Q_j^-(s', \cdot)$ is $L(s')$ -Lipschitz. For any reference μ_0 ,*

$$\|y_{\mu_1} - y_{\mu_0}\|_{L_2(\mathcal{D})} \leq \gamma \left(\mathbb{E}_{s'} L(s')^2 \cdot \|\mu_1(s') - \mu_0(s')\|_2^2 \right)^{1/2}. \quad (15)$$

If $L(s') \leq \bar{L}$, the right-hand side is at most $\gamma \bar{L} \sqrt{dD_{\text{critic}}}$. Conditional on μ_1 , later actors do not enter this one-step target perturbation.

Proposition 3 bounds a one-step value-target difference, not the stability of the learned critic. Its architectural content is that later refinement can alter D_{final} without directly entering this bound. With a Polyak target network, μ_1 is replaced by its target copy. A same-noise comparison under TD3 target smoothing obeys the same bound because action clipping is non-expansive. The current-state $\hat{D}_{\text{critic}}$ omits Polyak lag and target noise, whereas $\hat{D}_{\text{target}}$ measures the Polyak actor directly and has a smoothed counterpart. Neither displacement measures the action Lipschitz factor or the resulting change in critic value.

The separation introduces a complementary evaluation-policy mismatch: critic targets follow the μ_1 branch, whereas evaluation follows μ_K . Increasing D_{final} at fixed D_{critic} can therefore make the critic less representative of the evaluated policy. We interpret BAR as exposing a measurable target/deployment trade-off; its empirical return captures the net effect of target movement, evaluation mismatch, actor compute, and the coupled optimization trajectory.

4.4 Local Wasserstein-flow limit

The gradient-flow interpretation is a local organizing principle. Fix a critic and an interior action a_0 . Let $g = \nabla_a \hat{Q}(a_0)$, $H = \nabla_a^2 \hat{Q}(a_0)$, and $\beta = dh/C$. Before projection, the action-metric-matched linearized and exact statewise proximal updates satisfy

$$\Delta a_{\text{lin}} = \beta g, \quad \Delta a_{\text{prox}} = \beta \nabla_a \hat{Q}(a_0 + \Delta a_{\text{prox}}). \quad (16)$$

If the Hessian is locally Lipschitz and $\beta \|H\| < 1$, the latter yields

$$\Delta a_{\text{prox}} = \beta g + \beta^2 H g + O(\beta^3), \quad (17)$$

and both ideal updates are first-order discretizations of $\dot{a} = d\nabla_a \hat{Q}(a)$. With fixed C this is a constant time-rescaling of raw-critic gradient ascent; with a positive policy-dependent global C it follows the same direction under an adaptive time parameterization. With only C^2 regularity, the safe remainder is $o(\beta^2)$. The displayed expansion additionally requires small realized displacement, inactive projection, and an exact statewise solve. At large budgets, our experiments diagnose the resulting nonlocal actor-critic regime instead of extrapolating Eq. (17). The end-to-end return sweep, even at small nominal T , is not a frozen-critic action-level consistency test of this limit.

Proposition 4 (Local fixed-budget composition). *Let a smooth one-step update in local coordinates satisfy $\Psi_h(x) = x + hf(x) + h^2 p(x) + O(h^3)$. For fixed K and $T \rightarrow 0$,*

$$\begin{aligned} \Psi_{T/K}^K(x_0) &= x_0 + T f_0 \\ &\quad + T^2 \left[\frac{K-1}{2K} J_0 f_0 + \frac{1}{K} p_0 \right] + O(T^3). \end{aligned} \quad (18)$$

where $f_0 = f(x_0)$ and $J_0 = Df(x_0)$. Coordinate explicit Euler has $p_0 = 0$, whereas exact backward Euler has $p_0 = J_0 f_0$. Hence both are first-order methods; for each fixed K , the $T \rightarrow 0$ expansion has a leading flow-bias coefficient proportional to $1/K$.

Proposition 4 explains a possible fixed-critic small-step benefit of substepping. It neither predicts the sign of true-return change nor applies to the projected, saturated, moving-critic branches studied at large T . The expansion is not a uniform $K \rightarrow \infty$ result.

5 EXPERIMENTS

We evaluate BAR on the nine D4RL locomotion tasks formed by HalfCheetah, Hopper, and Walker2d with medium, medium-replay, and expert datasets (Fu et al., 2020). These tasks directly exercise deterministic continuous control and provide distinct behavior-data regimes for a target-routing stress test. Base implementations follow CORL configurations (Tarasov et al., 2022); scores are D4RL normalized returns. Additional per-task results and diagnostic definitions are in the Supplementary Material.

Fixed-budget protocol and estimand. We hold a total nominal budget T fixed, split it across K actor hops as $h = T/K$, and sweep

$$T \in \{0.05, 0.1, 0.2, 0.4, 0.7, 1.5, 2.5, 4, 7, 10, 12, 14, 17, 20\}.$$

over all nine tasks. The proximal grid uses seeds 0–3, for $9 \times 14 \times 4 = 504$ runs per depth; the linearized grid

uses seeds 0–1. Proximal seeds 0–1 and 2–3 were executed on separate hosts and form disjoint replication pairs. The grid is a high-resolution stability phase diagram rather than a final benchmark alone. For $K = 1$, Proposition 1 maps T to the normalized TD3+BC coefficient $\alpha = 2T$.

We compare TD3+BC ($K = 1$), proximal BAR with $K \in \{2, 3, 4\}$ (BAR-Prox), and an action-metric-matched projected linearized target with $K \in \{2, 3\}$ (BAR-Lin):

$$a_k^{\text{tar}} = \Pi_{\mathcal{A}} \left[\nu_k + d \frac{T}{K} \frac{\nabla_a \hat{Q}(s, \nu_k)}{C_{k,t}^L} \right], \quad (19)$$

$$\nu_1 = a, \quad \nu_k = \text{sg } \mu_{k-1,t}^+(s) \quad (k \geq 2).$$

At each hop, BAR-Lin detaches the reference, recomputes the $\hat{Q}$ gradient and scale, clips the target to the action box, and takes one mean-squared-error optimizer step with the corresponding persistent actor. Its first-hop scale is

$$C_{1,t}^L = \text{sg} \left(\mathbb{E}_{(s,a) \sim \mathcal{D}} |\hat{Q}_t(s, a)| + \epsilon \right),$$

whereas $C_{k,t}^L = C_{k,t}^P$ for $k \geq 2$. Thus the factor d matches the action metric, but the first-hop normalization point differs from BAR-Prox; the comparison does not match the complete scale schedule. It uses the same first-hop target and final-hop evaluation routing. All conditions match datasets, critic updates, and the 10^6 -update horizon; every headline score is the ten-episode D4RL normalized mean at the final checkpoint. Intermediate evaluation cadence varied by run family and does not enter this estimand. A K -hop condition performs K actor optimizer steps at each delayed policy update. The primary estimand is the end-to-end effect of depth-indexed BAR, which jointly changes local coefficient, re-centering frequency, actor compute, normalization trajectory, routing, and optimization trajectory.

The analyses test whether this bundle reduces large-budget low-return outcomes, whether the effect persists under the linearized realization, how first-hop target-policy displacement differs from final-policy reach, and which critic-validity diagnostics characterize failed training trajectories.

5.1 Additional actors broaden the stability envelope

Figure 2 and Table 1 summarize the fixed-budget sweep. The high-budget estimand averages seeds and the seven $T \geq 4$ budgets within each task, then weights all nine tasks equally. We descriptively call a seed-mean environment–budget cell collapsed when

Table 1: Aggregate fixed-budget results. High denotes $T \geq 4$ and tail denotes $T \geq 10$. Collapse entries are seed-mean cells / raw seeded runs; P and L denote proximal and linearized.

Method	Seeds	$T \leq 2.5$	High	Tail	Collapse
TD3+BC	4	66.68	25.38	20.27	34 / 142
BAR-P2	4	70.64	41.20	31.01	22 / 111
BAR-P3	4	70.53	51.92	42.23	16 / 84
BAR-P4	4	70.82	63.97	56.79	8 / 54
BAR-L2	2	69.02	36.68	27.44	29 / 58
BAR-L3	2	68.54	43.91	33.12	24 / 53

its score is below 20; this is a locomotion failure indicator, not a theoretical threshold. Across four-seed proximal $K = 1, 2, 3, 4$, the high-budget mean increases $25.38 \rightarrow 41.20 \rightarrow 51.92 \rightarrow 63.97$, while collapsed cells decrease $34 \rightarrow 22 \rightarrow 16 \rightarrow 8$ out of 63. Relative to TD3+BC, $K = 4$ gains 38.59 points with a task-resampling 95% interval of $[21.39, 54.63]$ and eight of nine positive task means. Relative to $K = 3$, it gains 12.06 points with interval $[3.84, 21.11]$ and seven of nine positive task means. Its cellwise median gain is only +.34: eight collapsed cells are rescued, none reverse, and those cells contribute 54.7% of the total added-depth gain. The stricter $T \geq 10$ tail is also strictly ordered, $20.27 \rightarrow 31.01 \rightarrow 42.23 \rightarrow 56.79$, so the depth ordering is not created by the $T = 4$ boundary. BAR-Lin with $K = 3$ raises the two-seed high-budget mean from 36.68 to 43.91 and reduces collapsed cells from 29 to 24. Thus refinement depth primarily rescues low-return outcomes rather than uniformly shifting settings.

Task dependence and replication. The budget response is smooth on HalfCheetah-medium, plateau-like on Hopper-medium-replay, sharply nonmonotone on Hopper-medium and Walker2d-medium, and asymmetric on expert datasets. There is no universal critic-valid radius. In a within-environment rank audit over the five diagnostic budgets, nominal T is the strongest absolute rank predictor of return in eight of nine environments, versus one for $\hat{D}_{\text{critic}}$ and none for $\hat{D}_{\text{final}}$; realized movement is therefore a mechanism readout, not a substitute for the randomized budget intervention. P4 minus P3 is positive after averaging within each dynamics family: +.79 for HalfCheetah, +13.04 for Hopper, and +22.33 for Walker2d. At Hopper-medium, $K = 4$ scores 100.9 ± 1.7 at $T = 17$ but becomes highly variable at $T = 20$ (41.8 ± 44.0), relocating rather than linearly extending a stability window. More broadly, both host-separated seed pairs reproduce the full high-budget ordering: seeds 0–1 give 25.97, 42.02, 52.78, 63.61, and seeds 2–3 give 24.79, 40.38, 51.05, 64.33 for $K = 1, 2, 3, 4$.

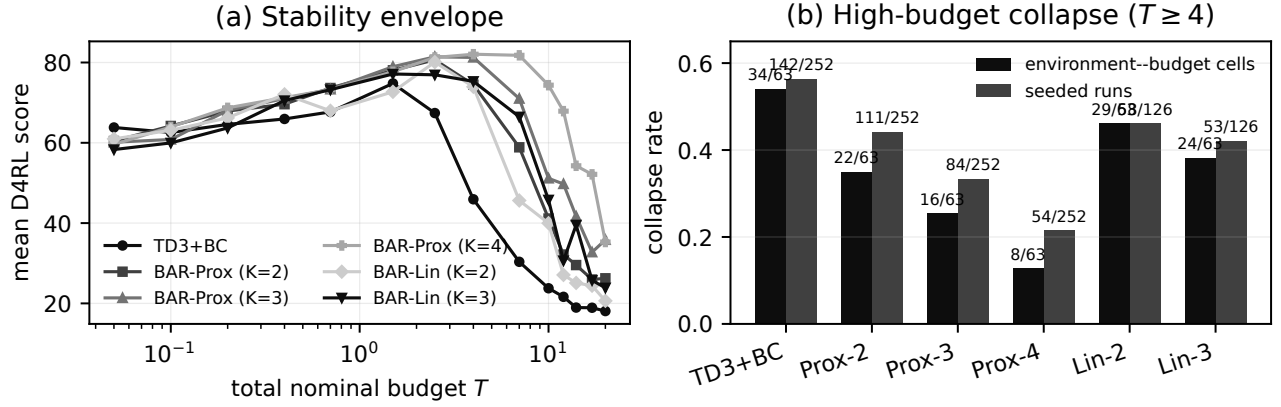


Figure 2: Fixed-budget sweep over nine locomotion tasks. (a) Mean score over tasks. (b) Collapse rates for $T \geq 4$, reported both by environment–budget cell and by seeded run. Proximal curves and cell means use four seeds; projected-linearized curves use two. Both realizations broaden the stable high-budget region with depth.

Their collapsed-cell sequences are 35, 25, 17, 9 and 35, 26, 16, 10. These are disjoint-seed replications of one method family, not independent implementations.

The linearized sweep shows that the gain is not unique to the proximal loss. At high budgets, BAR-Lin with $K = 3$ gains 7.23 points over $K = 2$ and reduces collapse cells from 29 to 24. BAR-Prox remains stronger overall, so these are comparisons of complete training procedures rather than isolated integrator effects.

5.2 Large-budget behavior is nonlocal and critic dependent

At $T = 20$, the projected BAR-Lin $K = 2$ target is clipped at an action bound in 94–95% of states, and the effective action-step coefficient h_{eff} is only 14% and 9% of the nominal dT/K coefficient in the two hops. These are medians over 18 environment–seed runs per hop. The practical operation is projected target generation followed by amortized neural regression under a moving critic. The sweep measures the resulting end-to-end stability envelope; local truncation error lies outside this estimand.

Figure 3 shows the common failure signature. Environment return supplies the external label: collapsed runs score below 20. These runs have orders-of-magnitude larger TD-error tails, larger critic magnitudes, and larger realized movements. The controlled common critic is the frozen final TD3+BC $T = 1$ critic matched by environment and seed. A run-total gain sums the mean Q change across actor hops on the shared state batch. For collapsed BAR-Lin runs, its median under the jointly trained critic is 3.33×10^{11} ; under the common critic it is -1.42 , compared with $+2.11$ for stable runs. For BAR-Prox, the common-

critic gain changes from $+2.51$ in stable runs to -1.06 in collapsed runs. This sign reversal diagnoses cross-critic disagreement consistent with Eq. (14); environment return remains the criterion for policy failure.

A simulator-reference audit provides targeted evidence of critic miscalibration in the screened checkpoints: the median signed error $\hat{Q}_1(s, a_K) - Q_{\text{MC}}^{\rho_1}(s, a_K)$ is $+1.70 \times 10^{12}$ in five collapsed checkpoints and -41.4 in 55 stable ones, using the median of checkpoint-level state medians. Episode-cluster intervals exclude zero for critic error at 58 of 60 checkpoints, and restoring the sampled episode clock preserves the error sign at all 60. Its action-ranking subset is sparse, so we use it only as targeted checkpoint-level evidence.

5.3 Target-action displacement and final-policy reach

At the same nominal budget, BAR-Prox realizes a smaller current-state first-hop displacement proxy than BAR-Lin once $T \geq 7$ (the Supplementary Material). At $T = 20$, their median $\hat{D}_{\text{critic}}$ values are 0.195 and 0.323. Across 90 matched environment–budget–seed cells at $T \in \{4, 7, 10, 14, 20\}$, exactly one of BAR-Prox and BAR-Lin collapses in 11 cases. In all 11, the stable realization has both smaller $\hat{D}_{\text{critic}}$ and smaller $\hat{D}_{\text{final}}$. These matched-cell outcomes associate smaller realized movement with the stable optimization trajectory. Because the proxy is measured on current validation states rather than the Polyak target policy on next states, this result organizes the observed failure transition without identifying a causal route through the TD target.

We measure Eq. (12) on all 270 audited method checkpoints. The current-state first-branch proxy closely

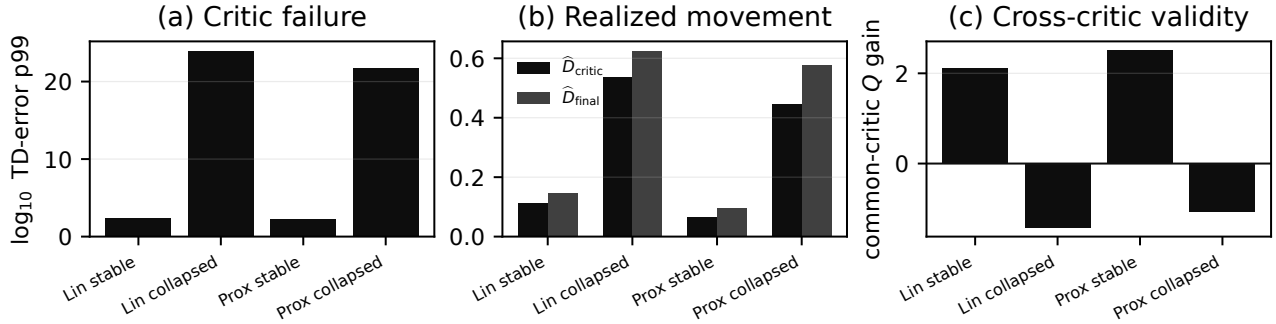


Figure 3: Median diagnostics over the matched $T \in \{4, 7, 10, 14, 20\}$ audit. Proximal and linearized realizations share a diagnostic pattern consistent with critic miscalibration: critic errors and realized movements increase sharply, and run-total gains under a common critic become negative. Environment return labels failure; the common critic diagnoses cross-critic disagreement.

tracks the deterministic target-action metric ($r = .9999$). BAR-Prox $K = 3$ has lower sample-anchored target-action displacement than TD3+BC in 88 of 90 matched cells (median ratio .658), while the current-state $\hat{D}_{final}$ proxy is lower in only 36 of 90 (median ratio 1.057). A few large final-policy contractions make the arithmetic mean difference negative despite this typical-cell expansion. Thus audited three-hop BAR exhibits the intended asymmetric movement response rather than merely shrinking both branches. Against $K = 2$, $K = 3$ lowers target-action displacement in 84 of 90 runs but loses by more than one score point in 17 of those 84, showing that exposure alone does not determine performance (the Supplementary Material).

A targeted Hopper-medium audit exhibits both forms of decoupling. Over $T = 4$ –10, $K = 3$ extends final-policy reach at nearly unchanged first-hop movement; near the $T = 12, 14$ transition it contracts both movements and avoids the low-return outcome. At $T = 20$, both depths saturate and fail (Supplement).

5.4 Targeted component controls

In eight post-hoc route-shadow runs, final-route calibration error is larger in all four-checkpoint aggregates when each shadow is evaluated under its native continuation (mean difference .124, sign-flip $p = .0078$). Under a common first-route continuation, the difference remains positive in six of eight runs but is inconclusive by the pre-specified rule (mean .116, $p = .164$). The shadows never drive the actors, so these are calibration diagnostics rather than return interventions. Compute-matched sequential, fixed-reference, and direct-final controls score 71.4, 70.0, and 59.4; slower $K = 1$ target averaging does not rescue selected cells. These controls favor routed multi-actor refinement over added optimizer calls or target lag alone

without isolating a single sweep-wide cause (Supplement).

5.5 Discussion and limitations

Both realizations reduce large-budget failures, with BAR-Prox stronger overall. The branch assignment separates exposure from reach by construction, and the audited $K = 3$ chain realizes the intended asymmetry: lower target-action displacement coexists with slightly greater typical-cell final-policy reach. The performance estimate is for the implemented depth intervention, in which K jointly changes local step, re-centering, compute, normalization, routing, and optimization trajectory. A saved-checkpoint actor-path audit reinforces this finite-step interpretation: frozen small-step maps compose locally, while the learned persistent endpoints remain far from frozen continuations and from one another across depths.

The proximal cross- K grid uses four seeds executed as two host-separated pairs; the linearized grid and mechanism audits use two. Exact per-run manifests for proximal seeds 2–3 were not retained, so same-procedure provenance rests on the archived launch configuration and score matrices rather than a configuration hash. The nine tasks span only three dynamics families, collapse is descriptive, and mechanism controls target post-hoc cells. We do not provide an offline rule for selecting T or K . Actor-update time and memory scale as $O(K)$ while critic updates and dataset sample source are fixed. The ideal fixed-critic analysis motivates, but does not certify, the live chain.

6 CONCLUSION

BAR makes refinement depth a practical control over offline actor-critic stability. By allocating a nomi-

nal finite-step coefficient budget across persistent actors, it routes only the first actor into Bellman targets while using a separately refined final actor for deployment, structurally separating which branch determines exposure and reach. Across four proximal seeds, the equally task-and-budget-weighted $T \geq 4$ mean is strictly ordered from 25.38 at $K = 1$ to 63.97 at $K = 4$, and low-return cells fall from 34/63 to 8/63; both disjoint host-separated seed pairs reproduce the aggregate ordering. The finite-step perturbation bound and 270-method-checkpoint TD3/P2/P3 audit identify bootstrap-exposure decoupling as the mechanism signature accompanying this expansion, with most of the added-depth gain coming from low-score rescue. These results establish K as BAR’s effective refinement-depth control and bootstrap-exposure decoupling as its observed mechanism signature.

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Checklist

- For all models and algorithms presented, check if you include:
 - A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes/No/Not Applicable]
Yes. Sections 3–4 define the setting, actor chain, losses, and assumptions.
 - An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes/No/Not Applicable]
Yes. Section 4 states that BAR uses K persistent actors and K actor optimizer calls per delayed update; the theoretical properties are stated in Propositions 2–4.
 - (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes/No/Not Applicable]
Yes. The supplementary material specifies the implementation and dependencies; the anonymized release contains launch, evaluation, diagnostic, and verification scripts.
- For any theoretical claim, check if you include:
 - Statements of the full set of assumptions of all theoretical results. [Yes/No/Not Applicable]

- Yes.** Each proposition states its assumptions, and the local-flow assumptions are separated from the coupled large-budget regime.
- (b) Complete proofs of all theoretical results. [Yes/No/Not Applicable]
Yes. Complete proofs are provided in the supplementary material.
- (c) Clear explanations of any assumptions. [Yes/No/Not Applicable]
Yes. Sections 4 and 5 explain the learned-critic, fixed- K , small-step, and implementation restrictions.
3. For all figures and tables that present empirical results, check if you include:
- (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes/No/Not Applicable]
Yes. Released CSVs, figure-generation code, a numerical claim verifier, and build instructions reproduce every main figure and aggregate table.
- (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes/No/Not Applicable]
No. All recoverable settings are listed, but exact per-run BAR-P4 configurations and manifests for proximal seeds 2–3 are unavailable.
- (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes/No/Not Applicable]
Yes. Captions and text define the equally task-and-budget-weighted high-budget estimand, seed-mean cells, raw seeded runs, collapse thresholds, paired ratios, and task-resampling intervals. The proximal grid uses four seeds in two host-separated pairs; the linearized grid and mechanism audits use two.
- (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes/No/Not Applicable]
No. The released artifact records software and execution protocols but does not contain a complete, independently verified inventory of all external-machine hardware.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
- (a) Citations of the creator If your work uses existing assets. [Yes/No/Not Applicable]
- Yes.** D4RL, TD3+BC, and the CORL implementation basis are cited.
- (b) The license information of the assets, if applicable. [Yes/No/Not Applicable]
Yes. The code release identifies its license; upstream dependencies retain their own licenses.
- (c) New assets either in the supplemental material or as a URL, if applicable. [Yes/No/Not Applicable]
Yes. An anonymized code and compact-results artifact accompanies the submission.
- (d) Information about consent from data providers/curators. [Yes/No/Not Applicable]
Not Applicable. We use public simulated-control benchmark datasets and collect no human data.
- (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes/No/Not Applicable]
Not Applicable. The assets contain simulated locomotion trajectories and no personal or sensitive content.
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
- (a) The full text of instructions given to participants and screenshots. [Yes/No/Not Applicable]
Not Applicable.
- (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Yes/No/Not Applicable]
Not Applicable.
- (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Yes/No/Not Applicable]
Not Applicable.